COURSE NOTES — BLOCK II: FINANCIAL MANAGEMENT & ACCOUNTING

Financial Management (I): Financial Statements & Capital Structure

Official Course Sources: Lecture Slides Topic 3 (Prof. Carmen Paz-Aparicio) & Practice Exercises T3 • UC3M Aerospace Engineering

1. Introduction: Economic Structure vs Financial Structure

The finance area of a company is the collection and management of the funds (capital) it uses. The finance manager faces two decisions: **investing decisions** (what to invest in and when?) and **financing decisions** (how to obtain the money to invest?) [Slide 3].

Economic Structure (Assets) [Slide 4]

"What is owned". Defined by the investments carried out by the company; it represents the use of the financial resources in terms of goods (assets) and rights.

Generates profitability [Slide 5].

Financial Structure (Equity + Liabilities) [Slide 4]

"What is owed". Defined by the sources of financing; it includes all the financial resources used to carry out the investments (equity and debt).

Carries a cost [Slide 5].

OBJECTIVE: MAXIMIZE VALUE [SLIDE 5, SLIDE 6]

$$\text{Profitability} > \text{Cost}$$

Investing decisions allocate capital to projects from which future profits are expected; companies invest in projects that create value (profits > costs). Financing decisions are the choice of equity or debt (internal equity or self-financing, external equity, or debt) and involve decisions about the capital structure and about dividends [Slide 6].

2. The Income Statement

The first statement prepared. It reports the business performance for a specific period; the simple format is $\text{Revenues} - \text{Expenses} = \text{Net Income}$ [Slide 7]. Revenues are earned when the sale of goods or services is made, and expenses are incurred when the business receives goods and services, whether or not the payment has been received or made [Slide 8] (the accrual principle).

Extended format [Slide 9]	Slide definition
Sales revenue	Revenues earned from the sale of goods or services (sales, service revenue, interest revenue) [Slide 8].
(-) Cost of goods sold	The expense incurred to buy or make a product for resale; a book bought for 25 USD and sold for 32 USD has COGS = 25 USD [Slide 10].
= Gross profit	Sales revenue − COGS.
(-) Operating expenses	The usual expenses of operating the business: salaries, utilities, depreciation [Slide 10].
= Income from operations	Gross profit − Operating expenses.
(+/-) Non-operating items	Revenues, expenses, gains and losses not related to the primary operations: interest expense, gains and losses on sale of equipment and investments [Slide 11].
= Income before taxes	Income from operations ± Non-operating items.
(-) Income taxes	Income before taxes times the tax rate; example: 50,000 at 30% gives 15,000 [Slide 11].
= Net income	Income before taxes − Income taxes.

Standard reference, not on the slides: net sales after discounts, the EBIT/EBT labels and the stock expression $\text{COGS} = \text{Initial stock} + \text{Purchases} - \text{Ending stock}$ are used in the problem sheets.

3. The Balance Sheet

The balance sheet reports the financial position of an accounting entity at a particular point in time [Slide 12]:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

Assets [Slide 13-15]

Economic resources owned by the company.

- **Current assets:** used or turned into cash within one year: cash, accounts receivable, inventory, short-term investments, supplies, prepaids.
- **Non-current assets:** the remainder: long-term investments, land, building, equipment, patents.

Liabilities and Equity [Slide 13, 16, 17]

- **Current liabilities:** obligations paid or satisfied within the coming year (accounts payable, unearned revenues).
- **Long-term liabilities:** obligations not paid or satisfied within the year (bonds payable).
- **Stockholders' equity:** contributed capital + retained earnings. Equity is the residual: assets minus liabilities.

LAYOUT [SLIDE 18]

$$\text{Current assets} + \text{Non-current assets} = \text{Current liabilities} + \text{Long-term liabilities} + \text{Stockholders' equity}$$

Worked example on the slides [Slide 19]

Data: cash 5,000; accounts receivable 10,000; inventories 45,000; supplies 4,000; buildings 65,000; accounts payable 12,000; bonds payable 40,000; common stock 45,000; additional paid-in capital 20,000; beginning retained earnings 5,000; sales 100,000; COGS 58,000; utility expense 8,000; supplies expense 3,000; salaries expense 16,000; interest expense 5,000; tax rate 30%. The slide gives no solution; the figures below are our own recomputation (checked in Python).

Step	Computation	Result
Gross profit	$100{,}000 - 58{,}000$	42,000
Operating expenses	$8{,}000 + 3{,}000 + 16{,}000$	27,000
Income from operations	$42{,}000 - 27{,}000$	15,000
Income before taxes	$15{,}000 - 5{,}000$	10,000
Income taxes	$0.30 \times 10{,}000$	3,000
Net income	$10{,}000 - 3{,}000$	7,000
Ending retained earnings	$5{,}000 + 7{,}000$	12,000
Total assets	$5{,}000 + 10{,}000 + 45{,}000 + 4{,}000 + 65{,}000$	129,000
Total liabilities	$12{,}000 + 40{,}000$	52,000
Total equity	$45{,}000 + 20{,}000 + 12{,}000$	77,000

Check: $129{,}000 = 52{,}000 + 77{,}000$.

Standard reference, not on the slides: accumulated depreciation as a contra-asset ($\text{NBV} = \text{Cost} - \text{Accumulated depreciation}$, straight-line $D = (\text{Cost} - \text{Residual})/n$), uncalled share capital and reserves appear in the problem sheets but are not defined on the Topic 3 slides.

4. Financing Decisions and the Financial Structure [Slide 20-35]

Business financing is the collection of the financial resources needed to develop the company's activity; their combination is the **capital (financial) structure** [Slide 20]. The slides split it into company resources (equity) and external resources (liabilities) [Slide 21, Slide 22].

Category	Instrument	Slide definition
Equity	**Self-financing** [Slide 23]	Retained profits (resources generated within the company). Profits are retained (self-financing) or distributed (dividends). Pay-out = Dividends/Profits; self-financing % = Retained earnings/Profits. Example: profits 100,000, dividends 75,000, retained 25,000 (pay-out 75%, self-financing 25%).
	External equity: shares [Slide 25, Slide 26]	Funds raised from owners at the start (initial capital) and by rights issues. Not paid back at any expiring date; a share can be converted into cash only by sale or transmission. A share is one of a finite number of equal portions of the capital, entitling to dividends and to a portion of the company's value in liquidation; dividends are not compulsory. Share values: nominal, accounting (book), intrinsic, market.
Long-term liabilities	**Bonds** [Slide 29]	Debt security: the issuer owes the holders a debt and must repay the principal and interest (the coupon) at maturity. Issuer = borrower, bondholder = lender.
	Long-term loans and bank credits [Slide 31]	A loan is a sum received that must be paid back at a certain time at a cost (interest). A bank credit makes a maximum amount available; interest is paid only on the amount borrowed (interest rates are higher).
	Leasing [Slide 33]	Use of fixed assets against contractual periodic payments; the lessee may buy the asset at the end. Types: finance and operating leasing (not further defined on the slide).
Short-term liabilities	**Accounts payable (trade credit)** [Slide 35]	Payment of goods is made some time after reception; suppliers offer the possibility to defer payment.
	Trade discount [Slide 35]	The financial institution makes a note (bill of exchange, promissory note, ...) available to the customer before expiry, after deducting expenses (interest, commission, ...).
	Factoring [Slide 35]	The company sells a customer credit to a factoring company before it expires; the factor handles reimbursement and assumes the insolvency risk.

5. Financial Leverage and Weighted Average Cost of Capital

Financial Structure Metrics [Slide 36]

Let E denote equity and L liabilities:

$$\%\ \text{Equity} = \frac{E}{E+L}, \qquad \%\ \text{Liabilities} = \frac{L}{E+L}, \qquad \text{Debt-to-Equity} = \frac{L}{E}$$

The debt-to-equity ratio evaluates financial leverage: the degree to which the company finances operations through debt versus wholly-owned funds. It reflects the ability of shareholder equity to cover all outstanding debts in a downturn; higher leverage ratios tend to indicate higher risk to shareholders. Slide example: $E +$

$L = 1{,}000$ M€, $E = 700$ M€, $L = 300$ M€, so $\%E = 70\%$, $\%L = 30\%$ and $L/E = 300/700 = 0.4286$.

Cost of Capital [Slide 37]

All financial resources have a cost: an **implicit (opportunity) cost** for equity and an **explicit cost** for liabilities. The capital cost is the weighted average cost of all the financial resources obtained by the company:

COURSE FORMULA [SLIDE 37]

$$\mathbf{K_{WACC} = K_{EQ} \cdot \left(\frac{E}{E + L}\right) + K_{L} \cdot \left(\frac{L}{E + L}\right)}$$

Standard reference, not on the slides: with tax-deductible interest the debt term becomes $K_L(1 - t)$ (tax shield). The slide formula has no tax term and is the one used in the course.